

Working with Henry Wynn: a challenge and an exceptional opportunity

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June 5, 2026

Abstract

Our dear friend and collaborator Henry P. Wynn passed away on 2 November 2024, at the age of 79. This paper highlights some of his contributions to statistics—particularly to experimental design—and acknowledges the profound positive impact his enthusiasm and generosity in sharing ideas have had on the community.

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HENRY P. WYNN (1945 – 2024)

1 Introduction

HENRY P. WYNN was involved in several major advances in experimental design and related areas in statistics. His connection with the mODa workshops has a long and rich history. He was among the initiators of the workshop series in 1987 and took an active part in nearly all subsequent mODa workshops (he missed only mODa 10 and 12). He was co-editor of the proceedings of the 3rd, 5th and 7th event, and organised mODa 7 in the Netherlands. Henry had several collaborators from the mODa community. In addition to the authors of this paper, we can mention Rainer Schwabe (see Section 3) and Werner Müller on Approximate Bayesian Computation Design (ABCD) [39, 40]. Without a doubt, his presence at past workshops is fondly remembered by all participants: his insightful comments, thoughtful questions, and warm encouragement of young researchers were always greatly appreciated. Henry’s presence at mODa will be profoundly missed.

Henry’s contributions to experimental design date back to his early papers [91, 92] on the sequential construction of D and D_s -optimal designs—works prepared during his PhD at Imperial College, under the supervision of Sir David Cox, that remain widely cited today. Later, another significant contribution,

following a sabbatical at Berkeley, is his joint paper with Jeff Wu [90], which provides a theoretical analysis of the convergence properties of general step-length algorithms for optimal design construction. At Berkeley Henry also befriended and collaborated with Jack Kiefer, yielding one of the earliest results on designs with correlated observations [46]. He later assisted in editing Kiefer’s collected works [45]. In the late 80’s Henry was part of the team led by Jerome Sacks that initiated the field of computer experiments; their influential paper [84] is arguably the most cited in the area.

At the very heart of Henry’s scientific pursuit was a deep concern for the foundations of the scientific method of learning from observed facts, as expounded in the philosophy of Sir Francis Bacon. He was attracted by the diptych active-vs-passive learning, namely passive observations and/or active learning through controlled experimentation. In [94] he states that “In general a mixture of controlled and random experimentation is best” (restated as a principle in [57]). He elaborated his viewpoint in a more mature form during a forgotten forum in Rome [95]; two of the present authors participated in the discussion. As late as 2023, in [62] he formulated his standing on passive vs controlled observations to introduce the algebraic circuits approach to A/B testing and randomised control trials. We regard the exploration of the passive/active divide, by statisticians and by science philosophers, as part of Henry’s legacy.

But Henry’s interests were certainly not limited to experiments. His interest in the application of information theory started with the introduction of Maximum Entropy Sampling [89] and remained throughout his career [87]. Still a PhD student, Henry published an important paper on multiple comparisons [98], a work that was later expanded through a long-standing collaboration with Daniel Naiman [58, 59], which involved ideas from computational geometry and topology. Tube theory considered in both papers permits one to derive tight inclusion-exclusion bounds for the probability of unions of events, which finds an application in system reliability [26, 28].

Order relations is another area of mathematics in a prominent position among Henry’s scientific curiosities (which may sound a little bizarre, given that he was not the most tidy and best organised person one could come across). Over a dozen of his papers have the word “ordering” in the heading and/or abstract, all the ones with Alessandra Giovagnoli (AG), but also with Robert Bohrer [13, 14] and Andrew Francis [24]. His collaboration with AG in this field is the object of Section 2.

Section 3 reviews the decade from 1994 to around 2004, a formative period during which the foundational ideas of Algebraic Statistics were established, and the theory of one-generator lattice designs for trigonometric models was developed, alongside several other projects that Henry pursued in parallel, many of them jointly with Eva Riccomagno (ER).

Henry’s work on design of experiments used his pioneering paper [66] as a stepping stone to continue in innovate ways. The presentation of Henry and Hugo Maruri-Aguilar’s (HMA) joint work in the period 2006 onwards (Section 4) is split into two main areas: (i) the study of direct design problems and saturated interpolators and their application to tensor grids; (ii) the regularisation and

smoothing of response surface models linking to spline-like interpolators for experimental designs.

The long-standing collaboration Henry maintained with Luc Pronzato (LP) and Anatoly Zhigljavsky (AZ) focussed on two distinct areas of research: (i) the applications of dynamical systems to search and optimisation, which led to a series of results that are surveyed in Section 5; (ii) the extension of the notion of variance, based on pairs of elements, to k -tuples, which led to a simplicial measure of dispersion that extends Wilks generalised variance for a multivariate distribution. Results are summarised in Section 6.

Henry had a remarkable ability to integrate theoretical work with practical applications. His motivation stemmed largely from the influence that experimental design, and statistical methods more generally, could exert in engineering contexts.

He played a pivotal role in the European Network for Business and Industrial Statistics (ENBIS). He served as its first President from 2000 to 2001 and was later honoured with the Box Medal in 2011. Henry took up the Chair in Mathematical Statistics at City University in 1985, later advancing to Dean of Mathematics and co-founding the Engineering Design Centre. During this time, he was instrumental in introducing Robust Engineering Design to Europe, inspired by Jerry Sacks's NSF workshops in the USA and the growing interest in quality improvement in both the USA and UK. Through a series of EU grants, Henry collaborated with major European companies such as Fiat (notably with Roberto Fontana at Centro Ricerche Fiat), SNECMA and Volvo. His work continued at the University of Warwick, where he became Professor of Industrial Statistics and founding Director of the Risk Initiative and Statistical Consultancy Unit (RISCU) in 1995. Risk theory was long one of his favourite subjects, as Henry was always deeply concerned with ethical issues. His later work [96] reflects his strongly humanistic outlook.

At RISCU, he secured further UK and European grants, expanding his focus to environmental modeling, particularly with the Swedish Environmental Research Institute (IVL). Under his leadership, RISCU became a leading UK group, undertaking numerous industrial projects. Throughout this period, Ronald Bates—who worked with Henry at City University, Warwick, and the London School of Economics—was his closest Research Fellow and later took a senior position at Rolls-Royce. Publications [47, 10, 4, 5] give a flavour of this activity.

Working with Henry was always a mixture of intensive brainstorming, fun and animated argumentation (to use a euphemism). Challenging discussions next to the blackboard alternated with intellectual treats with a cup of strong tea at breakfast or a pint of beer after dinner.

2 Henry Wynn and the quest for orderings

From the cradle onward, comparisons help our minds formalise concepts. Ideas like heat, weight, size, etc. come long before considering their measurements, and total or partial orderings are a straightforward consequence. This is even more relevant in statistics, with terms like *dispersion*, *variability*, *spread*, *scatter*, *peakedness*, *concentration*, *association*, *dependence* that remain vague if not properly defined through some order relation. In mathematics, order relations and order-preserving functions ($x < y$ implies $f(x) \leq f(y)$) are the key to proving inequalities. Henry Wynn’s scientific production shows his early interest in inequalities (see for instance [93]), hence his later exploration of so many facets and applications of orderings in statistics (see also Section 4.1).

2.1 The beginnings: DoE and majorisation

At the start, as already mentioned, Henry’s main scientific interest was Optimal Designs. The focus was on the search for the best experimental design for a given ANOVA model with respect to a specific “optimality” criterion (for instance D -optimality or A -optimality) related to minimising the variance of the LS estimates of the treatment effects. The designs that were found to perform better with respect to the most common criteria were the ones with some sort of “balance”. In 1976-77 Henry and AG—independently of each other—encountered majorisation when Ingram Olkin presented his then forthcoming book [48] at Imperial College in London. This was the beginning of their collaboration: Henry and AG introduced majorisation into DoE in a systematic way and this opened up a better understanding of optimality.

Majorisation is a partial ordering of n -dimensional real vectors: $\mathbf{y} \prec \mathbf{x}$ indicates that the entries of vector $\mathbf{y}$ are less spread out than $\mathbf{x}$. Its striking feature is that it can be defined in a wide variety of equivalent ways. One is through rearrangement of the entries: given $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ let $x_{(1)} \geq x_{(2)} \geq \dots \geq x_{(n)}$ and $y_{(1)} \geq y_{(2)} \geq \dots \geq y_{(n)}$ be the rearranged entries; $\mathbf{y} \prec \mathbf{x}$ iff

$$\sum_{i=1}^k x_{(i)} \geq \sum_{i=1}^k y_{(i)} \quad k = 1, \dots, n-1; \quad \sum_{i=1}^n x_i = \sum_{i=1}^n y_i. \quad (1)$$

The others are by means of sums $\sum_{i=1}^k u(x_i)$ with u a convex function, or by doubly stochastic matrices, or geometrically (Definition (1) means that $\mathbf{y}$ is in the convex hull of all transformations of $\mathbf{x}$ by permutations of its entries), and so forth; see [49]. The majorisation-preserving functions are named *Schur-convex*.

Removing the “equal means” condition of (1) defines two—perhaps less well-known—extensions:

- *Lower weak majorisation*: $\mathbf{y} \prec_w \mathbf{x}$ iff $\sum_{i=1}^k x_{(i)} \geq \sum_{i=1}^k y_{(i)} \quad k = 1, \dots, n$ ($\mathbf{y}$ is “smaller” and less spread out than $\mathbf{x}$); the order is preserved by Schur-convex functions that are non decreasing entrywise, after rearrangement.

- *Upper weak majorisation*: $\mathbf{y} \prec^w \mathbf{x}$ iff $\sum_{i=n-k}^n x_{(i)} \leq \sum_{i=n-k}^n y_{(i)}$ $k = 0, 1, \dots, n-1$ ($\mathbf{y}$ is “larger” and less spread out than $\mathbf{x}$); the order is preserved by non increasing Schur-convex functions.

It turns out that the commonly used design optimality criteria are non increasing Schur-convex functions of the ordered eigenvalues of the information matrix $\mathbf{M}$. For example, taking $u(x)$ to be either $-\log x$ or $1/x$ gives the common D - and A -optimality criteria respectively; thus upper weak majorisation of the $\mathbf{M}$ eigenvalues (in reverse) appears to be the natural ordering in design settings and the search for optimality translates into the search for designs whose information matrix has eigenvalues larger and less spread out than all the other ones.

In [31] it was proved that among designs with b blocks and given block sizes $k_1, \dots, k_b$ the Gale-Ryser one is the worst for estimating treatment contrasts, thus establishing a lower bound to the efficiency of designs in that class. In [32] and [34] majorisation showed the way to improve a given block design, either with or without a control treatment, leading to several optimality statements. It should be stressed that this was achieved by applying Kiefer’s continuous “approximation”: in this way properties were discovered that had been hidden in the literature by the strong emphasis on the combinatorial structure.

2.2 G for group, W for weak

Let us move on to 1983. AG was an academic visitor at Imperial College in that glorious summer, which was an unbounded sequence of drinking parties, but not very profitable for research. Henry was by no means a moralist, but come September tactfully put the question: was it not time to get some work done? At that point, like in a vision, GW -majorisation appeared before Alessandra’s eyes. She and Henry realised that the information matrices of designs can be compared directly, not just through their eigenvalues. The search for the most relevant matrix ordering lead to more insight into the already existing elegant theory of G -majorisation: the group of permutations is replaced by any linear group G acting on a real vector space. Replacing entrywise ordering with a general order relation $\triangleleft$ “compatible” with G —namely $\mathbf{y} \triangleleft \mathbf{x}$ implies $g(\mathbf{y}) \triangleleft g(\mathbf{x}) \forall g \in G$ —it was shown in [33] that two weak forms of G -majorisation could be introduced, *upper* and *lower GW -majorisation*. Several properties of GW -majorisation were proved for a general G and then specified to information matrices by taking subgroups of the orthogonal group acting by congruence on the space of n -dimensional semi-positive definite matrices and taking $\triangleleft$ to be the Loewner ordering. A not-so-anonymous referee of [33] painstakingly corrected 84 (yes!) minor errors in it, but said he liked the paper.

In [30] this way of comparing information matrices was applied to finding the best designs for linear regression models. It was proved that improving an experimental design by means of upper GW -majorisation leads, among other things, to improving the information for all subsets of estimable functions of the parameters.

In [36] G -majorisation was studied when G is the cyclic group on n elements. From the theory there stemmed an ordering on the smoothness of periodic functions with possible applications to time series analysis, and also an ordering of smoothing operators.

2.3 Stochastic orderings and dispersion

If orderings are fascinating in general, those of random quantities (stochastic orderings) are even more so. Stochastic orders have a long history. The one that almost monopolises statisticians' attention is *First order Stochastic Dominance* between two real random variables, $Y \leq^{st} X$ iff $F_Y(x) \geq F_X(x) \forall x$, which quantifies that X is “larger” than Y .

However, there are several other ways of comparing sizes of r.v.'s, and furthermore, apart from dominance, there are many more properties of r.v.'s that are captured by order relations, see [88, 56]. An example is dispersion. By the early nineties, Henry felt that in the multivariate case dispersion was still lacking a thorough investigation in terms of orderings. In wanting to study dispersion, Henry was motivated by the so-called “Taguchi methods” which had suddenly become popular in engineering in the previous decade, aimed at improving the stability of systems by reducing the dispersion of the output variable. So in [35] Henry together with AG defined a dispersion ordering between multivariate rv's $\mathbf{X}$ and $\mathbf{Y}$ in a strong ($\leq_{SD}$) and a weak ($\leq_D$) sense.

The strong version is $\mathbf{Y} \leq_{SD} \mathbf{X}$ when $\mathbf{Y}$ is stochastically equivalent to $\mathbf{k}(\mathbf{X})$, for some function $\mathbf{k}$ which is a *contraction* of $\mathbb{R}^n$, namely $\|\mathbf{k}(\mathbf{x}) - \mathbf{k}(\mathbf{x}')\| \leq \|\mathbf{x} - \mathbf{x}'\|$ for all $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$. The weaker ordering $\leq_D$ is defined by $\mathbf{Y} \leq_D \mathbf{X}$ iff $\|\mathbf{Y} - \mathbf{Y}'\| \leq^{st} \|\mathbf{X} - \mathbf{X}'\|$, where $\mathbf{Y}, \mathbf{Y}'$ are i.i.d. and so are $\mathbf{X}, \mathbf{X}'$; $\leq_{SD}$ implies $\leq_D$.

One characterisation of $\leq_D$ is in terms of the average probability content of balls with the same radius having centres distributed as $\mathbf{Y}$ and $\mathbf{X}$ respectively, another is through comparison of the L_2 norm of the respective probability density functions (i.e. the L_2 distances from the uniform distribution with the same support). For $\leq_{SD}$, the contraction property of $\mathbf{k}$ translates to a Jacobian condition, which makes it possible to compare $\leq_{SD}$ to other multivariate dispersion orderings in the literature. In particular, one type of stochastic majorisation defined in [48] as $\mathbf{Y} \sim^{st} D\mathbf{X}$, where D is a doubly stochastic matrix, turns out to be a special case of $\leq_{SD}$.

Both dispersion orders were studied for normally distributed $\mathbf{X}$ and $\mathbf{Y}$: there is a connection to majorisation via the covariance matrices. For the usual linear normal model, GW -majorisation of the information matrices implies the $\leq_D$ -ordering of the parameter estimators, which is thus stronger than the usual D -optimality.

Paper [35] lacks an in-depth study of the order-preserving functions of $\leq_D$ and $\leq_{SD}$. Years later, Henry together with Luc Pronzato and Anatoly Zhiglavsky, investigated measures of $\leq_{SD}$ that are extensions of Wilks generalised variance; see Section 6.

Henry was also concerned that most of the existing literature focused on real (continuous) random variables and there was less work on stochastic orderings for discrete ones. A contribution to the theory is [37], which shows that there are nice features specific to the discrete case that can be used for some closer analysis, in particular the use of Möbius inversion. Cone orderings and the algebra of Section 3.2 proved to be useful tools in this theory.

2.4 A duality theorem for some stochastic orderings

This is the story of an academic flop—one of the many that scatter the path of researchers’ life.

Henry was always genuinely interested in exchanging ideas with academics of different areas. In 2006 he was a Professor at LSE and visited AG at the School of Economics in Bologna. There is a rich Italian production in the studies of concentration of income, poverty, inequality, welfare, etc. with stochastic orderings as an attractive way of summarising preferences between distributions in collective decision-making. Stimulated by contacts with economics colleagues and personal political sympathies, Henry and AG were carried away into writing a joint paper ([38]) on the mathematics that links risk theory with social choice.

There are two different theories related to income distributions: a risk/utility one for individual preferences, and a social welfare one for collective ones. In the former, the key stochastic ordering is *Second order Stochastic Dominance* $Y \leq^{SSD} X$, defined as

$$\int_{-\infty}^x F_1(z) dz \leq \int_{-\infty}^x F_2(z) dz, \quad \forall x \in \mathbb{R}, \quad (2)$$

where $X \sim F_1$ and $Y \sim F_2$. First order Stochastic Dominance (see Section 2.3) holds if and only if $E(u(Y)) \leq E(u(X))$ for all non-decreasing $u(\cdot)$. For *SSD* the equivalence is for all non-increasing concave $u(\cdot)$. In the latter theory, the relevant order relation is the *Generalised Lorenz ordering* $\leq^{GL}$:

$$Y \leq^{GL} X \iff \int_0^p F_2^{-1}(\alpha) d\alpha \leq \int_0^p F_1^{-1}(\alpha) d\alpha, \quad \forall p \in [0, 1] \quad (3)$$

where F^{-1} is the quantile function. The order $\leq^{GL}$ is not the same as the usual *Lorenz ordering* which only compares inequality. The two theories are dual, thanks to the equivalence of (2) and (3), which acts as a “bridge”, transferring results from Expected Utility Theory and decision making under risk to results about social choice, with *inequality aversion* as the natural equivalent of *risk aversion*.

A generalisation of the equivalence of (2) and (3) is at the heart of [38]. Given two base *distortion* functions u_0 and v_0 :

- (i) $u_0 : \mathbb{R} \rightarrow \mathbb{R}$ is non decreasing and left continuous
- (ii) $v_0 : [0, 1] \rightarrow [0, 1]$ is non decreasing, right continuous, and $v_0(0) = 0$, $v_0(1^-) = 1$,

applied to F and to F^{-1} respectively, two orderings were defined in [38] and proved to be equivalent:

$$\begin{aligned} \text{(I)} \quad & \int_{-\infty}^x v_0(F_1(z)) \, du_0 \leq \int_{-\infty}^x v_0(F_2(z)) \, du_0 \quad \text{for all } x \text{ in } \mathbb{R}, \\ \text{(II)} \quad & \int_0^p u_0(F_1^{-1}(\alpha)) \, dv_0 \geq \int_0^p u_0(F_2^{-1}(\alpha)) \, dv_0 \quad \text{for all } p \text{ in } [0, 1]. \end{aligned}$$

The main result is that there are 8 equivalent statements of the above equivalence, for instance

$$\int_{-\infty}^{\infty} v_0(F_1(x)) \, du(x) \leq \int_{-\infty}^{\infty} v_0(F_2(x)) \, du(x)$$

for all non decreasing u_0 -concave functions u , and also

$$\int_0^1 u_0(F_1^{-1}(\alpha)) \, dv(\alpha) \geq \int_0^1 u_0(F_2^{-1}(\alpha)) \, dv(\alpha)$$

for all non decreasing v_0 -concave functions v . This shows that the “bridge” can be widened at least to an 8-lane one, to carry across a wider class of indicators, accounting for satisfaction, deprivation etc.

Paper [38] was submitted to an econometric journal, but one referee did not share the authors’ enthusiasm and the paper got rejected. A few years later it ended up in the Philosophy Papers of LSE and has peacefully fallen into quasi-oblivion ever since, while its authors moved on to pastures new.

3 The rise and coming of age of Algebraic Statistics

Henry was an exceptional mentor, although not for the weak spirited. During his last couple of years at City University, he employed a recent mathematics graduate from Italy as his research assistant (that is, ER). At the Engineering Design Center, among a whirlwind of rapid learning across a range of topics, their most successful project was on one-generator lattice designs for Fourier models, developed with Rainer Schwabe and Ron Bates and briefly referred to in Section 3.1. While changing jobs from City University to Warwick University, Henry found time with Giovanni Pistone to lay the foundation of a new field, now called Algebraic Statistics, in which a design is viewed as the zero set of a finite system of polynomial equations (see Section 4). One of Henry’s final grants at Warwick supported a major interdisciplinary collaboration between the Departments of Biology, Engineering, and Statistics. As part of this work, the (by then not so recent) mathematics graduate found herself visiting a cattle farm to collect data, in order to apply differential-algebraic techniques to the structural identifiability of biological rational polynomial models for multi-strain species SIR modelling [17]. About the cow project we shall write no more.

3.1 Low discrepancy sequences for Fourier regression

In a number of joint papers over a span of fifteen years, Henry worked on the theory of optimum orthogonal fractions for Fourier regression models based on integer lattice designs, providing principled alternatives to conventional product design [2, 83, 8, 9, 6]. Lattice designs offer scalability, making them suitable for complex, high-dimensional and large-scale systems. Their central result is that many optimality criteria for Fourier models—including D -, A -, and E -optimality—can be characterised and constructed using integration lattices and their duals.

As often with Henry after the idea there came a good name, the notation and an algorithm or at least a constructive method. An n -points one-generator lattice design in the hypercube $[0, 1)^d$ is a uniform design supported on the equally spaced grid $L_{g,n}$ where $g = (g_1, \dots, g_d)$ is an integer valued vector with positive entries, $g_1 = 1$, and such that the greatest common divisor of its components and the sample size n is one. The grid is defined as

$$L_{g,n} = \left\{ \left(\frac{jg_1 \bmod n}{n}, \dots, \frac{jg_d \bmod n}{n} \right) : j = 0, \dots, n-1 \right\}.$$

While $F(d; m_1, \dots, m_d; M)$ denotes the complete Fourier model with *marginal* models of order $m_1, \dots, m_d$, and all *interactions* terms up to order M , defined as

$$\begin{aligned} \mathbb{E}(Y(x_1, \dots, x_d)) &= \theta_0 \\ &+ \sqrt{2} \sum_{l=1}^M \sum_{k_1 < \dots < k_l} \sum_{r_1=1}^{m_{k_1}} \dots \sum_{r_l=1}^{m_{k_l}} \sin(2\pi(\beta_1 r_1 x_{k_1} + \dots + \beta_l r_l x_{k_l})) \theta_{k_1 \dots k_l, r_1 \dots r_l, \beta} \\ &+ \sqrt{2} \sum_{l=1}^M \sum_{k_1 < \dots < k_l} \sum_{r_1=1}^{m_{k_1}} \dots \sum_{r_l=1}^{m_{k_l}} \cos(2\pi(\beta_1 r_1 x_{k_1} + \dots + \beta_l r_l x_{k_l})) \phi_{k_1 \dots k_l, r_1 \dots r_l, \beta}. \end{aligned}$$

The vector g generates a finite number of line segments in $[0, 1)^d$, which form a single closed line wrapped around the torus identified by matching opposite faces of the hypercube. The key observation is that $F(d; m_1, \dots, m_d; M)$ can be mapped into a one dimensional model with the same Fisher information matrix by exploiting the structure of the lattice. Orthogonality conditions are expressed in terms of the dual lattice, which gives information about the hyperplanes of dimension $d-1$ associated with $L_{g,n}$

$$h^\top x = k \text{ with } k \in \mathbb{Z} \text{ and } x \in L_{g,n}.$$

Theorem 2 in [83] gives necessary, sufficient, computable and algebraic conditions on g and $m_1, \dots, m_d$ so that the information matrix is orthogonal and the sample size n is minimal. This was called the power law, resonating with the Nyquist frequency. Notably n increases polynomially with d and M . Essentially, optimal designs correspond to lattices whose duals avoid low-frequency

aliasing, expressed in term of low order polynomial equations. Thus, dual lattice theory is used to analyse aliasing, orthogonality, and variance properties, providing clear geometric and algebraic conditions for optimality.

3.2 Algebraic Statistics and the GROSTAT workshops

Two other notable names coined by Henry are the title of the seminal paper [66] and GROSTAT, the series of workshops held from 1998 to 2003 on applications of computational commutative algebra to statistics, with emphasis on the design and analysis of experiments. In [66] Henry began a reinterpretation of statistical experimental designs using computational commutative algebra and polynomial ideals. Rather than treating designs merely as combinatorial objects, the authors of [66] consider a design as the zero locus of a set of polynomial equations. The set of all such polynomials forms a polynomial ideal (or design ideal), which is finitely generated and encodes essential information about the design structure. This algebraic representation offers several advantages:

- **Estimability and Identifiability:** algorithms based, for example, on Gröbner bases allow one to derive (all) sets of monomial terms (and hence model term selection) that are identifiable given any design supported on a finite set of distinct points. This reframes classical concerns (which terms in a polynomial regression model can be estimated without bias) in algebraic terminology using quotient vector spaces and a computer algebra software.
- **Aliasing and Confounding:** traditional definitions of aliasing—where multiple polynomial or interaction effects cannot be distinguished—are expressed as polynomial relations in the design ideal. The Gröbner basis systematically reveals these confounding structures and can extend classical factorial design theory.

Monograph [65], besides establishing the name of the newly born field of Algebraic Statistics, codifies the conceptual framework, establishes a common language, systematically develops this theory and its implications for experimental design, reliability analysis (see Section 4) and the broader concept of algebraic statistical models, by which statistical objects correspond to algebraic varieties. The algebraic perspective, pioneered by Henry in e.g. [27, 42, 64, 29, 50], enriches traditional statistical design theory by bridging combinatorial design theory with computational algebra, providing algorithmic tools for model exploration and identifiability, and extending classical ideas of confounding and estimability to richer design contexts.

Building on these seminal works, which provide practical tools for the practising statistician, Henry—together with a wide network of collaborators—went on to make further contributions to experimental design and to related areas such as reliability [54] and coherent systems [43]. In particular, the GROSTAT workshops fostered the involvement of a large group of Italian researchers, including Maria Piera Rogantin, Fabio Rapallo, and Roberto Fontana, who worked closely with Henry on the application of algebraic statistics to the design of

experiments, for examples on fractional factorial designs [63], aberration criteria [22, 11], saturated fractions [21], and robust design [23].

4 Algebra for designs

This section looks at the work by Henry and HMA on direct design problems following [65, 66]. It also looks at regularisation of over-parameterised models after Henry’s seminal work on emulation [84] and on industrial methods [4, 97]. The section later reviews reliability work by Henry and coauthors, after [58].

To set notation, an experimental design $\mathcal{D}$ in d factors is a set of n distinct points in $\mathbb{R}^d$; each point’s coordinates specify covariates levels determining an experimental condition; at the i -th design point there is a continuous observation y_i . Those responses are modelled with a polynomial function

$$y(\mathbf{x}) = \sum_{\alpha \in L} \theta_{\alpha} \mathbf{x}^{\alpha}.$$

The exponent vector α has non-negative integer entries $\alpha_1, \alpha_2, \dots, \alpha_d$; for indeterminates $x_1, x_2, \dots, x_d$ a monomial is $\mathbf{x}^{\alpha} := x_1^{\alpha_1} x_2^{\alpha_2} \dots x_d^{\alpha_d}$ and θ_{α} is the coefficient of $\mathbf{x}^{\alpha}$. The function $y(\mathbf{x})$ is also written $y(\mathbf{x}) = \mathbf{f}(\mathbf{x})^{\top} \boldsymbol{\theta}$ using vectors $\boldsymbol{\theta} = (\theta_{\alpha} : \alpha \in L)^{\top}$ and $\mathbf{f}(\mathbf{x}) = (\mathbf{x}^{\alpha} : \alpha \in L)^{\top}$.

The list L of distinct exponents gives structure to $y(\mathbf{x})$, and is referred to as the *model*. Consider hierarchical models, where for every monomial $\mathbf{x}^{\alpha}, \alpha \in L$, all its monomial divisors have exponents in L . The hierarchical set L is a well known object in combinatorics under the different names of staircases, Ferrers diagrams, integer partitions and Young tableaux. These hierarchical structures provide a flexible modelling framework in our statistical work.

For estimation of $\boldsymbol{\theta}$ collect y_i values in vector $\mathbf{Y}$. The y_i are not assumed to be functions of $\mathbf{x}$, thus this is empirical model building. The linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}$ requires the design model matrix $\mathbf{X}$ which is the evaluation of model terms at design points $\mathbf{X} = (\mathbf{s}^{\alpha})_{s \in \mathcal{D}, \alpha \in L}$ and $\mathbf{X}$ is assumed to have the largest possible rank, i.e. $\text{rank}(\mathbf{X}) = \min(n, \#L)$. Identifiability of the model is the ability to uniquely estimate the parameter vector $\boldsymbol{\theta}$ using data and a main concern is the search for identifiable models for a given design.

Three cases appear, depending on the dimensions of $\mathbf{X}$. When $\mathbf{X}$ has more rows than columns, this is linear regression widely studied in the literature with estimated $\boldsymbol{\theta}$ minimising the sum of squares of the error $\text{SSE}(\mathbf{Y}, \boldsymbol{\theta})$, see [55]. The following two sections survey work in the saturated case with square $\mathbf{X}$, and regularisation of the supersaturated $\mathbf{X}$ with more columns than rows.

4.1 Saturated models

With square $\mathbf{X}$ the model interpolates, i.e. $y(\mathbf{x}_i) = y_i$ for $i = 1, \dots, n$. The model is identifiable by invertible $\mathbf{X}$ with parameter estimate $\hat{\boldsymbol{\theta}} = \mathbf{X}^{-1}\mathbf{Y}$.

An arbitrary hierarchical L is not necessarily identifiable by $\mathcal{D}$ and to solve this, algebraic work by Henry and colleagues [66] identifies models for a design.

$$\mathbf{X} = \begin{pmatrix} 1 & x_1 & x_2 & x_1^2 & x_1x_2 & x_2^2 & x_1^3 & \cdots & x_2^3 & \cdots \\ 1 & -1 & -1 & 1 & 1 & 1 & -1 & \cdots & -1 & \cdots \\ 1 & 1 & -1 & 1 & -1 & 1 & 1 & \cdots & -1 & \cdots \\ 1 & -1 & 1 & 1 & -1 & 1 & -1 & \cdots & 1 & \cdots \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & \cdots & 1 & \cdots \end{pmatrix}$$

Table 1: Extended design model matrix for column selection in Example 1.

The *design ideal* $I(\mathcal{D})$ encodes the confounding relations between model terms in a design $\mathcal{D}$ and allows identification of hierarchical models for $\mathcal{D}$. Model identification requires a *term order* $\prec$ to compute the *reduced Gröbner basis* for $I(\mathcal{D})$ from where an identifiable model for $\mathcal{D}$ can be read. Besides the technical need, setting $\prec$ allows users' term preferences. The following example provides an illustration; one may also refer to [the] monograph [65].

Example 1 Consider $\mathcal{D} = \{(\pm 1, \pm 1)\}$, a 2^2 factorial design with ± 1 levels. As point set, $\mathcal{D}$ is the solution of the system $\mathcal{G} = \{x_1^2 - 1, x_2^2 - 1\}$. The polynomials in $\mathcal{G}$ generate the design ideal $I(\mathcal{D})$ and it can be shown that, for any term order, $\mathcal{G}$ is also a reduced Gröbner basis for $I(\mathcal{D})$. The leading terms of the Gröbner basis are x_1^2, x_2^2 and an identifiable model are those monomials not divisible by these: $1, x_1, x_2, x_1x_2$, equivalently we read $L = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$.

The algebraic method is equal to selecting the first n linearly independent columns in an extended design model matrix, after reordering columns according to $\prec$, see [1]. Table 1 shows columns of extended $\mathbf{X}$ for Example 1, ordered according to degree and $x_1 \prec x_2$. Besides identifying $\{1, x_1, x_2, x_1x_2\}$, aliasing relations can be read with algebra, e.g. in Table 1, $x_2^3 \equiv x_2$ over $\mathcal{D}$.

The method can be applied to many designs, like fractional factorial, response surface, screening and mixture designs, Latin squares and hypercubes, see [27, 29, 52, 65]. The analysis of fractional factorial designs with indicator functions is an important avenue of research, see [20, 19, 23].

Model identification with algebra depends on the chosen order $\prec$. The collection of models identified using all term orders is called the algebraic fan of the design, which can be expensive to compute, with software available only for moderate size computations [25]. The algebraic fan is a subset of the statistical fan, i.e. the set of all identifiable hierarchical models [15] which can be linked to and computed with recent Gröbner free algebra work [16].

Three types of designs are of interest regarding the fan: maximal fan, generic and minimal fan designs. Maximal fan designs identify all possible staircases, while a generic design identifies *corner cut* models. Corner cuts are those staircases that can be separated from their complement by a hyperplane, e.g. $\{1, x_1, x_2, x_1^2\}$ is corner cut while $\{1, x_1, x_2, x_1x_2\}$ is not. Importantly, the algebraic fan of a generic design is the set of corner cut models [61, 11].

A random design is maximal fan with probability one [65], yet its algebraic fan is the set of corner cuts which is generally much smaller. For generic designs,

Henry and coworkers developed a generalised notion of model complexity termed model aberration and found bounds for it [12, 11]. The pair design-model is linked to a corresponding pair when the design is embedded in a larger grid, for which the generalised aberration bounds can be used [50].

Generic designs can be studied using the power series of the complement of model L . This converging power series is the *Hilbert series* of $I(\mathcal{D})$, computed for a given $\prec$. This series is $(x_1^2 + x_2^2 - x_1^2 x_2^2)/((1 - x_1)(1 - x_2))$ for the model of Example 1, while it is $(x_1^3 + x_1 x_2 + x_2^2 - x_1 x_2^2 - x_1^3 x_2)/((1 - x_1)(1 - x_2))$ for the corner cut cited above. For a generic design, the models in the algebraic fan lead to a special Hilbert series computed for *lex-segment* ideals that have maximal complexity hence the model has minimal complexity [51].

Minimal fan designs identify only one model. Factorial designs, sparse and Smolyak grids are all echelon designs which are minimal fan. An echelon design is a design that, after coordinate relabelling, can be made into a hierarchical L . For echelon $\mathcal{D}$ when L has two extreme corners α, β , the interpolator satisfies inclusion-exclusion (IE) of smaller interpolators $y(\mathbf{x}) = y_\alpha(\mathbf{x}) + y_\beta(\mathbf{x}) - y_{\alpha \wedge \beta}(\mathbf{x})$ with $\alpha \wedge \beta := \min(\alpha, \beta)$ coordinatewise, see [53]. The inverse of $\mathbf{X}$ uses IE of generalised inverses $\mathbf{X}^{-1} = \mathbf{X}_\alpha^- + \mathbf{X}_\beta^- - \mathbf{X}_{\alpha \wedge \beta}^-$, so in echelon, three objects are linked by IE: $y(\mathbf{x})$, $\mathbf{X}^{-1}$ and the Hilbert series of the complement of L .

4.2 Regularising the supersaturated model

A supersaturated $\mathbf{X}$ does not lead to a unique estimate; to regularise use penalty $P(\mathbf{f}, \boldsymbol{\theta}) = \int_{\mathcal{X}} \text{tr}(H)^2 d\mathbf{x}$ [4] with Hessian $H = H(y(\mathbf{x})) = H(\mathbf{f}(\mathbf{x})^\top \boldsymbol{\theta})$ and $\mathcal{X}$ a compact subset of $\mathbb{R}^d$. A polynomial approximation to an interpolating spline is the smooth supersaturated model (SSM) [7]. The SSM is well suited for small size problems, does not use spline knots and works in complex regions [18]. To estimate $\boldsymbol{\theta}$ in SSM, response values are interpolated while minimising roughness:

$$\min_{\boldsymbol{\theta}} P(\mathbf{f}, \boldsymbol{\theta}) \text{ such that } \mathbf{X}\boldsymbol{\theta} = \mathbf{Y}.$$

As often with Henry, SSM work did not start in this final form and began as an optimisation in Gröbner basis computation adding dummy design points. Soon it became clear that estimation was a constrained quadratic optimisation problem with $P(\mathbf{f}, \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{K} \boldsymbol{\theta}$ and matrix $\mathbf{K} := \sum_{i,j} \int_{\mathcal{X}} \mathbf{f}^{ij} \mathbf{f}^{ij\top} d\mathbf{x}$ using partial derivatives $\mathbf{f}^{ij}$ with respect to x_i and x_j for $i, j = 1, \dots, d$. Also, hierarchical L implies closure of $\mathbf{f}$ with respect to derivatives with linear form $\frac{\partial \mathbf{f}}{\partial x_i} = \mathbf{D}_i \mathbf{f}$ and cross-derivative $\mathbf{f}^{ij} = \mathbf{D}_{ij} \mathbf{f}$ and $\mathbf{D}_{ij} = \mathbf{D}_i \mathbf{D}_j$. An equivalence theorem for roughness optimality is stated in [3], using rewritten $\mathbf{K} = \sum_{i,j} \mathbf{D}_{ij} \mathbf{M} \mathbf{D}_{ij}^\top$ and moment matrix $\mathbf{M} = \int_{\mathcal{X}} \mathbf{f}(\mathbf{x}) \mathbf{f}(\mathbf{x})^\top d\mathbf{x}$. A future alternative to SSM estimation is a Tikhonov type of regularisation by roughness

$$\min_{\boldsymbol{\theta}} (\text{SSE}(\mathbf{Y}, \boldsymbol{\theta}) + \lambda P(\mathbf{f}, \boldsymbol{\theta})).$$

The regularised estimate is $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{K})^{-1} \mathbf{X}^\top \mathbf{Y}$, akin to ridge regression.

4.3 Algebra for reliability

Henry's interest in inclusion-exclusion (IE) led to reliability work with abstract tubes [58, 59, 60] which he later linked to his algebra work [26, 28].

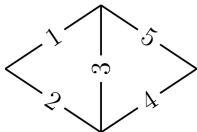


Figure 1: Network functioning if there is a connected path from left to right.

Figure 1 motivates this with a simple network. For simplicity, refer to network events by numbers, so e.g. **45** stands for the failure event of nodes labelled four and five. The Gröbner basis of the network failure ideal $\{\mathbf{12}, \mathbf{45}, \mathbf{134}, \mathbf{235}\}$ is the minimal set of events causing the failure of the network in the figure. This basis has already eliminated the redundant generator first provided **1245**.

By associating independent failure probabilities to nodes, the Hilbert series of the ideal helps constructing Bonferroni bounds for the network reliability. Even for moderate size systems, naïve IE is expensive and algebra helps retrieve compact and efficient IE, see [85, 86] and more recently [54, 43]. In the example, the naïve approach to series using five generators then all their pairs, triads and so on has 31 terms, while the efficient approach has only 10 terms.

5 Dynamical systems in search and optimisation

This topic was the starting point for the collaboration between Henry, LP and AZ in the early 1990's. Book [72] contains the essentials of the results obtained from 1990 to 2000; [67] and [75] provide an overview of its content. The central idea is to view an optimisation algorithm, or a root-finding algorithm, as a dynamical system whose characteristics can be studied in detail. This dynamical system is obtained by renormalising the algorithm, or the search space, in an appropriate manner at each iteration. Thus, instead of having a fixed point $\mathbf{x}$ in $\mathbb{R}^d$ (the point where the function is extremal or zero), and iterations (or a sequence of regions) converging towards this point, a sequence of points $\mathbf{x}_i$ is obtained, corresponding to renormalised $\mathbf{x}$, which evolve in a fixed region of space. It is the transformation $\mathbf{x}_i \rightarrow \mathbf{x}_{i+1}$ that defines the dynamical system. To fully understand how it works, it took as many three-dimensional constructions on the sandy beach of Antibes as two-dimensional drawings on the blackboard.

The results primarily concerned the case $d = 1$, with a particularly comprehensive analysis of second-order line-search, i.e., Golden-Section (GS) type algorithms, that minimise a uniextremal function over an interval by using, at each iteration, two evaluations of the function in the current interval. A fundamental result of this work is that certain algorithms considered optimal (Fibonacci, GS) are in fact optimal for worst-case scenarios corresponding to a

set of zero measure for the invariant measure of the associated dynamical system. It is thus possible to do better on average, this average being calculated along a typical trajectory of the algorithm, for almost any initial point.

A general second-order line search algorithm for the minimisation of a function f having a unique minimiser x^* in an interval $\mathcal{I}_0 = [A_0, B_0]$ works as follows. Let E_0 be a point in $\mathcal{I}_0$; the algorithm is defined by $e_0 = (E_0 - A_0)/(B_0 - A_0)$ and a function $\phi : [0, 1] \rightarrow [0, 1]$. Let $\mathcal{I}_n = [A_n, B_n]$ be the current uncertainty interval (containing x^*) at iteration $n = 0, 1, \dots$. At the beginning of each iteration, $\mathcal{I}_n$ already contains one point E_n . Denote $L_n = B_n - A_n$, $e_n = (E_n - A_n)/L_n$ (which belongs to $[0, 1]$), define $e'_n = \phi(e_n)$, and $E'_n = A_n + e'_n L_n$, $U_n = \min\{E_n, E'_n\}$, $V_n = \max\{E_n, E'_n\}$. The updating rule for E_n and $\mathcal{I}_n$ only uses comparisons of function values at U_n and V_n :

$$\{E_{n+1}, \mathcal{I}_{n+1}\} = \begin{cases} \{V_n, [U_n, B_n]\} & \text{if } f(U_n) \geq f(V_n), \\ \{U_n, [A_n, V_n]\} & \text{otherwise.} \end{cases}$$

Every such second-order algorithm defines a dynamical system for (x_n, e_n) with $x_n = (x_* - A_n)/L_n$. Different definitions of the convergence rate of the algorithm translate into various exponents and entropies of the associated dynamical system. For example, the Rényi entropy of order 2 arises naturally as a measure of contraction of $\mathcal{I}_n$; see [69]. The famous GS algorithm corresponds to $e_0 \in \{\varphi, 1 - \varphi\}$ and $\phi(t) = 1 - t$, with $\varphi = (\sqrt{5} - 1)/2 \simeq 0.6180$, so that $e_n, e'_n \in \{\varphi, 1 - \varphi\}$ for all n . [The] paper [68] presents a general class of algorithms for which e_n remains in a finite set of cardinality q ; algorithms with $q = 2$ and $q = 3$ are fully described. For any $q \geq 4$ there are infinitely many constructions. The one in [70] is a special instance with $q = 4$ for which the asymptotic convergence rate is much faster than for the GS algorithm. The GS algorithm is symmetric, in the sense that $\phi(t) = 1 - t$. This class of algorithms is studied in [73]. A direct implementation of any symmetric algorithm has sub-exponential convergence due to numerical inaccuracies. The GS algorithm should be implemented by enforcing $e'_n \in \{\varphi, 1 - \varphi\}$; it has the fastest asymptotic convergence rate among all symmetric algorithms. The algorithm presented in [71] uses $\phi(t) = t + w$ if $t < 1/2$ and $\phi(t) = t - w$ otherwise. A suitable choice of w and an appropriate initialisation make the algorithm significantly faster than GS both asymptotically and for finite n when the function is locally symmetric at x^* .

The case $d > 1$ is much more difficult. On the one hand, significant achievements for ellipsoidal algorithms could not be obtained despite considerable effort [72, Chap. 6], and the dynamical-system point of view has little practical impact on these algorithms. On the other hand, the study of gradient-type algorithms for the minimisation of a quadratic function $f(\mathbf{x}) = (1/2) \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{x}^\top \mathbf{y}$ in $\mathbb{R}^d$, with $\mathbf{A}$ a symmetric positive-definite matrix and $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^d$, or A an operator and x and y in a Hilbert space, has led to fruitful developments. In [74] the methodology is applied to the steepest-descent algorithm and a generalisation of Akaike's famous result to the Hilbert-space case is obtained: the renormalised algorithm converges to a two-point attractor. This result is extended in [76] to a wider family of algorithms, containing in particular that of minimum residues.

Paper [77] concerns the s -gradient algorithm, which consists of repeating series of s iterations of the conjugate-gradient method, each one initialised with a steepest-descent iteration. The asymptotic behaviour is much more complex, the case $s = 2$ is studied in details. In particular, as the 2-gradient algorithm converges in one iteration in $\mathbb{R}^2$ and the steepest-descent trajectory belongs asymptotically to a two-dimensional space, alternating the two methods yields much faster convergence than each of them applied separately. In [41], various gradient-type algorithms are investigated, which are directly related to experimental design algorithms.

6 Simplicial measures of multivariate dispersion

This joint work between Henry, LP and AZ began by drawing triangles on the blackboard and asking: what would be the average area of all possible triangles formed from a cloud of n points? When the points are i.i.d. with μ , what does this tell us about μ ? How does this extend to simplices when $d > 2$? At the beginning, drawing many triangles did not help much... Let μ be a probability measure on $\mathbb{R}^d$ with mean E_μ and covariance matrix V_μ , and assume that V_μ is finite. We have

$$\int \int \|\mathbf{x}_1 - \mathbf{x}_2\|^2 \mu(d\mathbf{x}_1) \mu(d\mathbf{x}_2) = 2 \int \|\mathbf{x} - E_\mu\|^2 \mu(d\mathbf{x}) = 2 \text{trace}(V_\mu). \quad (4)$$

The central idea of this work is to extend the property above based on pairs of points to $(k + 1)$ -tuples of points, by considering the measure of dispersion

$$\phi_{[k],\delta}(\mu) = \int \dots \int \mathcal{V}_k^\delta(\mathbf{x}_1, \dots, \mathbf{x}_{k+1}) \mu(d\mathbf{x}_1) \dots \mu(d\mathbf{x}_{k+1}), \quad k \geq 2,$$

for some δ in $\mathbb{R}^+$, where $\mathcal{V}_k(\mathbf{x}_1, \dots, \mathbf{x}_{k+1})$ is the volume of the k -dimensional simplex (its area when $k = 2$) formed by the $k + 1$ vertices $\mathbf{x}_1, \dots, \mathbf{x}_{k+1}$ in $\mathbb{R}^d$, with $k = d$ as a special case.

Paper [79] concerns the case $\delta = 2$. Wilks generalised variance is a particular case of this, as $\phi_{[d],2}(\mu) = [(d + 1)/d!] \det(V_\mu)$. The case $k = 1$ corresponds to (4). For a general k , with the help of the celebrated Binet-Cauchy formula, the simplicial k -variance $\phi_{[k],2}(\mu)$ can be expressed as

$$\phi_{[k],2}(\mu) = \Phi_{[k],2}(V_\mu) = [(k + 1)/k!] \mathcal{E}_k(V_\mu),$$

with $\mathcal{E}_k(V_\mu)$ the elementary symmetric function of degree k of the d eigenvalues of V_μ . The concavity and differentiability of the functional $\phi_{[k],2}^{1/k}(\cdot)$ yields a necessary and sufficient condition for a measure μ^* to have maximum dispersion in the sense of $\phi_{[k],2}$. Moreover, the function $\mathbf{M} \mapsto \Phi_{[k],2}^{-1/k}(\mathbf{M}^{-1})$ is shown to be concave on the set of $d \times d$ symmetric positive definite matrices (so that $\Phi_{[k],2}(\mathbf{M}^{-1})$ is convex), which provides an equivalence theorem, i.e., a necessary and sufficient condition for the optimality of a design measure ξ when $\mathbf{M} = \mathbf{M}(\xi)$ is an information matrix.

Some of these results have been extended in [78] which establishes properties of measures maximising $\phi_{[k],\delta}(\mu)$ for general $\delta \geq 0$ (and minimising $\phi_{[k],\delta}(\mu)$ when $\delta < 0$).

In [80], the concept of simplicial k -variance is used to define an extension of Mahalanobis distance and a simplicial distance between distributions based on the Jeffreys–Bregman divergence for the functional $\phi_{[k],2}(\cdot)$. These distances are sensitive to the dimensionality of the data and hence are of particular interest for measures whose covariance matrices are close to singularity.

In addition to Jeffreys–Bregman, other divergences and distances between distributions are considered in [81] (which was presented at mODa 12), including the Kullback–Leibler, Burbea–Rao and Jensen–Shannon divergences and the Bhattacharyya distance. A general construction is given based on defining a directional derivative of a given concave functional, at one distribution in the direction of the other, and concavity, or strict concavity, is shown to influence the properties of the resulting divergence. For normal distributions, these divergences can be expressed in terms of means and covariances. Optimality criteria from experimental design (Kiefer’s φ_p class [44]) provide a family of functionals, sufficient conditions are given to ensure that they can distinguish normal distributions. The k -th order simplicial distance based on $\phi_{[k],2}(\cdot)$ is revisited from this aspect and is shown to be of practical interest when the distributions are close to being singular. The problem of choosing a suitable k is addressed in the paper.

7 Final remarks

Working with Henry P. Wynn was a privilege and a joy. Henry was a dedicated scholar, generous with his ideas, a leading author, although, it must be said, not easily persuaded to abandon a line of thought—even when he was wrong. However, should our account suggest that Henry thought of nothing but work, we wish to state most definitely that this is not an accurate picture. He certainly enjoyed his work, but also loved spending time with friends, good food, and a leisurely life. The conference organised in Juan-les-Pins (France) in 2005 to celebrate Henry’s 60th birthday was a perfect illustration of this balance between intellectual intensity and leisurely conviviality, drawing together a remarkable and far-reaching network of collaborators. Volume [82] published after the conference contains several papers surveying areas of his research and stands as evidence that hard work and pleasure need not be mutually exclusive.

We are grateful for the opportunity to contribute to this paper and to recall both the delights and the occasional exasperation of collaborating with Henry as a colleague and a friend. His passing was a profound loss to his family, friends and the scientific community, and his work remains a monument to a great scholar and a delightful human being.

References

- [1] E. Babson, S. Onn, and R. Thomas. The Hilbert zonotope and a polynomial time algorithm for universal Gröbner bases. *Advances in Applied Mathematics*, 30(3):529–544, 2003.
- [2] R.A Bates, R.J. Buck, E. Riccomagno, R. Schwabe, and H.P. Wynn. Experimental design and observation for large systems. *Journal of the Royal Statistical Society. Series B. Methodological*, 58(1):77–94, 95–111, 1996. With discussion and a reply by the authors.
- [3] R.A. Bates, P.R. Curtis, H. Maruri-Aguilar, and H.P. Wynn. Optimal design for smooth supersaturated models. *Journal of Statistical Planning and Inference*, 154:3–11, 2014.
- [4] R.A. Bates, B. Giglio, and H.P. Wynn. A global selection procedure for polynomial interpolators. *Technometrics*, 45(3):246–255, 2003.
- [5] R.A. Bates, R.S. Kenett, D.M. Steinberg, and H.P. Wynn. Achieving robust design from computer simulations. *Quality Technology and Quantitative Management*, 3(2):161–177, 2006.
- [6] R.A. Bates, H. Maruri-Aguilar, E. Riccomagno, R. Schwabe, and H.P. Wynn. Self-avoiding generating sequences for Fourier lattice designs. In *Algebraic methods in statistics and probability II*, volume 516 of *Contemp. Math.*, pages 37–47. Amer. Math. Soc., Providence, RI, 2010.
- [7] R.A. Bates, H. Maruri-Aguilar, and H.P. Wynn. Smooth supersaturated models. *Journal of Statistical Computation and Simulation*, 84(11):2453–2464, 2014.
- [8] R.A. Bates, E. Riccomagno, E. Schwabe, and H.P. Wynn. The use of lattices in the design of high-dimensional experiments. In *New developments and applications in experimental design (Seattle, WA, 1997)*, volume 34 of *IMS Lecture Notes Monogr. Ser.*, pages 27–35. Inst. Math. Statist., Hayward, CA, 1998.
- [9] R.A. Bates, E. Riccomagno, R. Schwabe, and H.P. Wynn. Lattices and dual lattices in optimal experimental design for Fourier models. *Computational Statistics & Data Analysis*, 28(3):283–296, 1998.
- [10] R.A. Bates, H.P. Wynn, R. Fontana, C. Randazzo, and E. Vaccarino. Empirical modelling of diesel engine performance for robust engineering design. In S.P. Edwards, D.M. Grove, and H.P. Wynn, editors, *Statistics for Engine Optimization*, pages 163–174. Wiley, 2000.
- [11] Y. Bernstein, H. Maruri-Aguilar, S. Onn, E. Riccomagno, and H.P. Wynn. Minimal average degree aberration and the state polytope for experimental designs. *Annals of the Institute of Statistical Mathematics*, 62:673–698, 2010.

- [12] Y. Berstein, J. Lee, H. Maruri-Aguilar, S. Onn, E. Riccomagno, R. Weismantel, and H.P. Wynn. Nonlinear matroid optimization and experimental design. *SIAM Journal on Discrete Mathematics*, 22(3):901–919, 2008.
- [13] R. Bohrer and H.P. Wynn. Spherically symmetric probability orderings useful in multiple comparisons. *The Annals of Statistics*, 10(4):1253–1260, 1982.
- [14] R. Bohrer and H.P. Wynn. Probabilistic ordering of Scheffé polyhedra. In *Inequalities in Statistics and Probability*, volume 5, pages 150–156. Institute of Mathematical Statistics, 1984.
- [15] M. Caboara, G. Pistone, E. Riccomagno, and H.P. Wynn. The fan of an experimental design. SCU Research Report 33, Department of Statistics, University of Warwick, May 1997.
- [16] M. Ceria, S. Lundqvist, and T. Mora. Degrobnerization: a political manifesto. *Applicable Algebra in Engineering, Communication and Computing*, 33(6):675–723, 2022.
- [17] M.J. Chappell, G. Margaria, E. Riccomagno, and H.P. Wynn. Differential algebra methods for the study of the structural identifiability of biological rational polynomial models. *Mathematical Biosciences*, 174:1–26, 2001.
- [18] P. Curtis and H. Maruri-Aguilar. Design for smooth models over complex regions. In J. Kunert, C.H. Müller, and A.C. Atkinson, editors, *mODa 11 - Advances in Model-Oriented Design and Analysis*, pages 71–78. Springer, 2016.
- [19] R. Fontana and G. Pistone. Algebraic generation of orthogonal fractional factorial designs. In N. Torelli, F. Pesarin, and A. Bar-Hen, editors, *Advances in Theoretical and Applied Statistics*, pages 61–71. Springer, Berlin, Heidelberg, 2013.
- [20] R. Fontana, G. Pistone, and M.-P. Rogantin. Classification of two-level factorial fractions. *Journal of Statistical Planning and Inference*, 87(1):149–172, 2000.
- [21] R. Fontana, F. Rapallo, and M.-P. Rogantin. A characterization of saturated designs for factorial experiments. *Journal of Statistical Planning and Inference*, 147:204–211, 2014.
- [22] R. Fontana, F. Rapallo, and M.-P. Rogantin. Aberration in qualitative multilevel designs. *Journal of Statistical Planning and Inference*, 174:1–10, 2016.
- [23] R. Fontana, F. Rapallo, and H.P. Wynn. Circuits for robust designs. *Statistical Papers*, 63(5):1537–1560, 2022.

- [24] A.R. Francis and H.P. Wynn. Subgroup majorization. *Linear Algebra and its Applications*, 444:53–66, 2014.
- [25] K. Fukuda, A.N. Jensen, and R.R. Thomas. Computing Gröbner fans. *Mathematics of Computation*, 76(260):2189–2212, 2007.
- [26] B. Giglio, D.Q. Naiman, and H.P. Wynn. Gröbner bases, abstract tubes, and inclusion-exclusion reliability bounds. *IEEE Transactions on Reliability*, 51(3):358–366, 2002.
- [27] B. Giglio, E. Riccomagno, and H.P. Wynn. Gröbner basis strategies in regression. *Journal of Applied Statistics*, 27(7):923–938, 2000.
- [28] B. Giglio and H.P. Wynn. Monomial ideals and the scarf complex for coherent systems in reliability theory. *The Annals of Statistics*, 32(3):1289–1311, 2004.
- [29] B. Giglio, H.P. Wynn, and E. Riccomagno. Gröbner basis methods in mixture experiments and generalisations. In *Optimum design 2000 (Cardiff)*, volume 51 of *Nonconvex Optim. Appl.*, pages 33–44. Kluwer Acad. Publ., Dordrecht, 2001.
- [30] A. Giovagnoli, F. Pukelsheim, and H.P. Wynn. Group-invariant orderings and experimental designs. *Journal of Statistical Planning and Inference*, 17(2):159–171, 1987.
- [31] A. Giovagnoli and H.P. Wynn. A majorization theorem for the C-matrices of binary designs. *Journal of Statistical Planning and Inference*, 4(2):145–154, 1980.
- [32] A. Giovagnoli and H.P. Wynn. Optimum continuous block designs. *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences*, 377(1771):405–416, 1981.
- [33] A. Giovagnoli and H.P. Wynn. G-majorization with applications to matrix orderings. *Linear Algebra and its Applications*, 67:111–135, 1985.
- [34] A. Giovagnoli and H.P. Wynn. Schur-optimal continuous block designs for treatments with a control. In *Proceedings of the Berkeley Conference in Honor of Jerzy Neyman and Jack Kiefer*, volume 1, pages 418–433. Wadsworth California, 1985.
- [35] A. Giovagnoli and H.P. Wynn. Multivariate dispersion orderings. *Statistics & Probability Letters*, 22(4):325–332, 1995.
- [36] A. Giovagnoli and H.P. Wynn. Cyclic majorization and smoothing operators. *Linear Algebra and its Applications*, 239:215–225, 1996.
- [37] A. Giovagnoli and H.P. Wynn. Stochastic orderings for discrete random variables. *Statistics & Probability Letters*, 78(16):2827–2835, 2008.

- [38] A. Giovagnoli and H.P. Wynn. (U,V)-ordering and a duality theorem for risk aversion and Lorenz-type orderings. *LSE Philosophy Papers, Centre for Philosophy of Natural and Social Science*, 2012.
- [39] M. Hainy, W.G. Müller, and H.P. Wynn. Approximate Bayesian Computation Design (ABCD), an introduction. In D. Uciński, A.C. Atkinson, and M. Patan, editors, *mODa 10—Advances in Model-Oriented Design and Analysis*, pages 135–143. Springer, 2013.
- [40] M. Hainy, W.G. Müller, and H.P. Wynn. Learning functions and Approximate Bayesian Computation Design: ABCD. *Entropy*, 16:4353–4374, 2014.
- [41] R. Haycroft, L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Studying convergence of gradient algorithms via optimal experimental design theory. In L. Pronzato and A. Zhigljavsky, editors, *Optimal Design and Related Areas in Optimization and Statistics*, chapter 2, pages 13–37. Springer, 2009.
- [42] T. Holliday, G. Pistone, E. Riccomagno, and H.P. Wynn. The application of computational algebraic geometry to the analysis of designed experiments: a case study. *Computational Statistics*, 14(2):213–231, 1999.
- [43] R. Iglesias, F. Mohammadi, P. Pascual-Ortigosa, E. Sáenz-de Cabezón, and H.P. Wynn. Stable coherent systems. *Applicable Algebra in Engineering, Communication and Computing*, 36(6):1267–1293, 2025.
- [44] J. Kiefer. General equivalence theory for optimum designs (approximate theory). *The Annals of Statistics*, 2(5):849–879, 1974.
- [45] J. Kiefer. *Collected Papers III. Design of Experiments*. Springer, 1985. L. Brown, I. Olkin, J. Sacks and H.P. Wynn, eds.
- [46] J. Kiefer and H.P. Wynn. Optimum balanced block and Latin square designs for correlated observations. *The Annals of Statistics*, 9(4):737–757, 1981.
- [47] N. Logothetis and H.P. Wynn. *Quality Through Design: Experimental Design, Off-Line Quality Control, and Taguchi’s Contributions*. Oxford University Press, 1989.
- [48] A.W. Marshall and I. Olkin. *Inequalities: Theory of Majorization and its Applications*. Academic Press, NY, 1979.
- [49] A.W. Marshall, I. Olkin, and B.C. Arnold. *Inequalities: Theory of Majorization and its Applications (second edition)*. Springer, NY, 2010.
- [50] H. Maruri-Aguilar, E. Sáenz-de Cabezón, and H.P. Wynn. Alexander duality in experimental designs. *Annals of the Institute of Statistical Mathematics*, 65(4):667–686, 2013.

- [51] H. Maruri-Aguilar, E. Sáenz-de Cabezón, and H.P. Wynn. Betti numbers of polynomial hierarchical models for experimental designs. *Annals of Mathematics and Artificial Intelligence*, 64(4):411–426, 2012.
- [52] H. Maruri-Aguilar and H.P. Wynn. The algebraic method in experimental design. In A. Dean, M. Morris, J. Stufken, and D. Bingham, editors, *CRC Handbook of Design and Analysis of Experiments*. CRC Press, 2015.
- [53] H. Maruri-Aguilar and H.P. Wynn. Sparse polynomial prediction. *Statistical Papers*, 64(4):1233–1249, 2023.
- [54] F. Mohammadi, E. Sáenz-de Cabezón, and H.P. Wynn. Types of signature analysis in reliability based on Hilbert series. *Journal of Symbolic Computation*, 79:140–155, 2017.
- [55] D.C. Montgomery, E.A. Peck, and G.G. Vining. *Introduction to Linear Regression Analysis*. Wiley, Hoboken, New Jersey, 6th edition, 2021.
- [56] A. Müller and D. Stoyan. *Comparison Methods for Stochastic Models and Risks*. John Wiley & Sons, 2002.
- [57] W.G. Müller and H.P. Wynn. Controlled versus “random” experiments: a principle. In J. Kunert, C.H. Müller, and A.C. Atkinson, editors, *mODa 11-Advances in Model-Oriented Design and Analysis*, pages 191–198. Springer, 2016.
- [58] D.Q. Naiman and H.P. Wynn. Inclusion-exclusion-Bonferroni identities and inequalities for discrete tube-like problems via Euler characteristics. *The Annals of Statistics*, pages 43–76, 1992.
- [59] D.Q. Naiman and H.P. Wynn. Abstract tubes, improved inclusion-exclusion identities and inequalities and importance sampling. *The Annals of Statistics*, 25(5):1954–1983, 1997.
- [60] D.Q. Naiman and H.P. Wynn. Improved inclusion-exclusion inequalities for simplex and orthant arrangements. *Journal of Inequalities in Pure & Applied Mathematics [electronic only]*, 2(2):Paper No. 18, 16 p.–Paper No. 18, 16 p., 2001.
- [61] S. Onn and B. Sturmfels. Cutting corners. *Advances in Applied Mathematics*, 23(1):29–48, 1999.
- [62] E. Pesce, F. Rapallo, E. Riccomagno, and H.P. Wynn. Generation of all randomizations using circuits. *Annals of the Institute of Statistical Mathematics*, 75(4):683–704, 2023.
- [63] G. Pistone, E. Riccomagno, and M.-P. Rogantin. Methods in algebraic statistics for the design of experiments. In *Optimal design and related areas in optimization and vstatistics*, volume 28 of *Springer Optim. Appl.*, pages 97–132. Springer, New York, 2009.

- [64] G. Pistone, E. Riccomagno, and H.P. Wynn. Gröbner basis methods for structuring and analysing complex industrial experiments. *International Journal of Reliability, Quality and Safety Engineering*, 7(04):285–300, 2000.
- [65] G. Pistone, E. Riccomagno, and H.P. Wynn. *Algebraic Statistics*. Chapman & Hall, Boca Raton, 2001.
- [66] G. Pistone and H.P. Wynn. Generalised confounding with Gröbner bases. *Biometrika*, 83(3):653–666, 1996.
- [67] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Dynamic systems in search and optimisation. *CWI Quarterly*, 8(3):201–236, 1995.
- [68] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Stochastic analysis of convergence via dynamic representation for a class of line-search algorithms. *Combinatorics, Probability & Computing*, 6:205–229, 1997.
- [69] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Using Rényi entropies to measure uncertainty in search problems’. In G.G. Yin and Q. Zhang, editors, *Proc. 1996 AMS–SIAM Summer Seminar, Lectures in Applied Math, Mathematics of Stochastic Manufacturing Systems, vol. 33*, pages 253–268, Providence, Rhode Island, 1997. American Math. Soc.
- [70] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. A generalised golden-section algorithm for line-search. *IMA Journal of Mathematical Control and Information*, 15:185–214, 1998.
- [71] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Finite sample behaviour of an ergodically fast line-search algorithm. *Computational Optimization and Applications*, 14:75–86, 1999.
- [72] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. *Dynamical Search*. Chapman & Hall/CRC, Boca Raton, 2000.
- [73] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Analysis of performance of symmetric second-order line search algorithms through continued fractions. *IMA Journal of Mathematical Control and Information*, 18:281–296, 2001.
- [74] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Renormalised steepest descent in Hilbert space converges to a two-point attractor. *Acta Applicandae Mathematicae*, 67:1–18, 2001.
- [75] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. An introduction to dynamical search. In P.M. Pardalos and H.E. Romeijn, editors, *Handbook of Global Optimization*, volume 2, chapter 4, pages 115–150. Kluwer, Dordrecht, 2002.
- [76] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Asymptotic behaviour of a family of gradient algorithms in $\mathbb{R}^d$ and Hilbert spaces. *Mathematical Programming*, A107:409–438, 2006.

- [77] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. A dynamical-system analysis of the optimum s -gradient algorithm. In L. Pronzato and A. Zhigljavsky, editors, *Optimal Design and Related Areas in Optimization and Statistics*, chapter 3, pages 37–80. Springer, 2009.
- [78] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Extremal measures maximizing functionals based on simplicial volumes. *Statistical Papers*, 57(4):1059–1075, 2016.
- [79] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Extended generalised variances, with applications. *Bernoulli*, 23(4A):2617–2642, 2017.
- [80] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Simplicial variances, potentials and Mahalanobis distances. *Journal of Multivariate Analysis*, 168:276–289, 2018.
- [81] L. Pronzato, H.P. Wynn, and A. Zhigljavsky. Bregman divergences based on optimal design criteria and simplicial measures of dispersion. *Statistical Papers*, 60(2):195–214, 2019.
- [82] L. Pronzato and A. Zhigljavsky, editors. *Optimal Design and Related Areas in Optimization and Statistics*. Springer, Heidelberg, 2009.
- [83] E. Riccomagno, R. Schwabe, and H.P. Wynn. Lattice-based D -optimum design for Fourier regression. *The Annals of Statistics*, 25(6):2313–2327, 1997.
- [84] J. Sacks, W.J. Welch, T.J. Mitchell, and H.P. Wynn. Design and analysis of computer experiments. *Statistical Science*, 4(4):409–435, 1989.
- [85] E. Sáenz-de Cabezón and H.P. Wynn. Betti numbers and minimal free resolutions for multi-state system reliability bounds. *Journal of Symbolic Computation*, 44(9):1311–1325, 2009. Effective Methods in Algebraic Geometry.
- [86] E. Sáenz-de Cabezón and H.P. Wynn. *Algebraic Reliability Based on Monomial Ideals: A Review*, pages 314–335. World Scientific, New Jersey, 2012.
- [87] P. Sebastiani and H.P. Wynn. Maximum entropy sampling and optimal Bayesian experimental design. *Journal of the Royal Statistical Society*, B62(1):145–157, 2000.
- [88] M. Shaked and J.G. Shanthikumar. *Stochastic Orders*. Springer, New York, NY, 2007.
- [89] M.C. Shewry and H.P. Wynn. Maximum entropy sampling. *Journal of Applied Statistics*, 14(2):165–170, 1987.
- [90] C.F.J. Wu and H.P. Wynn. The convergence of general step-length algorithms for regular optimum design criteria. *The Annals of Statistics*, 6(6):1273–1285, 1978.

- [91] H.P. Wynn. The sequential generation of D -optimum experimental designs. *The Annals of Mathematical Statistics*, 41:1655–1664, 1970.
- [92] H.P. Wynn. Results in the theory and construction of D -optimum experimental designs. *Journal of the Royal Statistical Society*, B34:133–147, 1972.
- [93] H.P. Wynn. An inequality for certain bivariate probability integrals. *Biometrika*, 64(2):411–414, 1977.
- [94] H.P. Wynn. Controlled versus random experimentation. *Journal of the Royal Statistical Society*, D31(3):237–244, 1982.
- [95] H.P. Wynn. The foundation of experimental design and observation. *Journal of the Italian Statistical Society*, 2(2):137—191, 1993. (with discussion).
- [96] H.P. Wynn. *Against Sacrifice: an Essay on Risk and Ethics*. Matador, San Jose, California, USA, 2021.
- [97] H.P. Wynn and R.A. Bates. Emulator technology in engineering design. *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture*, 213(3):305–309, 1999.
- [98] H.P. Wynn and P. Bloomfield. Simultaneous confidence bands in regression analysis. *Journal of the Royal Statistical Society*, B33(2):202–217, 1971.